MALAYSIA SOVEREIGN AI & DIGITAL INFRASTRUCTURE COMPREHENSIVE REPORT

Published: October 2026 | **Frameworks Covered:** MDEC MDLR, MyIPO Act 834, Sovereign Cloud Nodes

1. Executive Summary & Documented Project Posture

This official compilation document synthesizes the strategic positioning of Malaysia's sovereign digital transformation, specifically focusing on decentralized AI computing networks, regional innovation nodes, and nationwide intellectual property alignments. Based on the initial filing framework, key deployments under the core engine **SGAE 2.0** and regional nodes highlight an expansion of Tier-1 connectivity extending from centralized national infrastructures down to strategic satellite nodes.

Project/Registry Name	Filing Code & Status	Infrastructure Scope & Focus
SAZWAN DIGITAL VALLEY Sungai Petani, Kedah	GI-2026-07734 Pending Registration KDH-SDV-2026-001	Primary innovation hub for SGAE 2.0 core engine development and decentralized identity (DID) research.
KEDAH AI NEXUS Northern Corridor, MY	GI-2026-07735 Pending Registration KDH-KAN-2026-002	Strategic gateway for AI services in the Northern Corridor, hosting sovereign cloud nodes and KMS integrations.

2. National Policy Framework Alignment

• **MyIPO Geographical Indications Act 2022 (Act 834):** Secures legal territorial exclusivity for technologies originating from specified locations like Sungai Petani, linking digital innovations to localized sovereignty.
• **MDEC Malaysia Digital Location Recognition (MDLR):** Grants strategic validation to spaces hosting Tier-1 Nodes, paving the way for exclusive Malaysia Digital (MD) corporate status, fiscal incentives, and rapid tech expat onboarding.
• **SHA-256 Kriptografi Digest:** Serving as an immutable cryptographic timestamp within the MyIPO registry to secure source code integrity.

3. Nationwide Comprehensive Ecosystem Breakdown (By State)

Region / State	Core Infrastructure Focus & Assets
Kuala Lumpur	Central Command & National Hub. Houses MDEC HQ (Bangsar South), Oracle Sovereign AI Cloud ($6.5B investment), Bitdeer AI Neocloud, and massive MD Hub presence (WORQ, Common Ground).
Selangor	Cyberjaya Hyperscale Tier-3/Tier-4 Data Center cluster. Houses NEXTDC KL1 (65MW Tier IV certified flagship), Autonomous Vehicle (AV) test beds, and drone technology sandboxes.
Kedah	AI R&D; (Sungai Petani Hub), Kulim Hi-Tech Park (KHTP) Tier-1 Advanced Semiconductor Fabrication, and Bukit Kayu Hitam Border Connectivity Gateway (D8-1 Hub).
Johor	Fastest growing Hyperscale Data Center corridor in Southeast Asia. Powered by sustainable energy networks at Sedenak Tech Park (STPi) and YTL Green Data Center Park.
Pulau Pinang	The 'Silicon Valley of the East'. Strategic center for integrated circuit (IC) Design, global semiconductor packaging, and subsea cable landing stations.
Sarawak	Sarawak Digital Economy Blueprint. Focuses on regional green-energy powered data centers, sovereign digital identity frameworks, and localized startup ecosystem building.
Sabah	BIMP-EAGA regional cross-border e-commerce hub, rural broadband infrastructure deployments, and geographical mapping intelligence nodes.
Pahang	Kuantan East Coast international subsea cable landing facility, serving as a low-latency direct gateway to the South China Sea network matrix.
Perak	Digital Perak creative and animation infrastructure (Ipoh/Meru hub) along with precision medical hardware digital components factories.
Negeri Sembilan	Malaysia Vision Valley 2.0 (MVV 2.0) advanced smart logistics networks and ancillary disaster recovery backup compute facilities.
Melaka	National cybersecurity resilience sandboxes and regional smart heritage/tourism digitalization software nodes.
Terengganu	Industrial IoT applications integration for offshore energy extraction and localized SME e-commerce enablement ecosystems.
Perlis	Smart Agriculture AI research initiatives powered by UniMAP and Chuping Valley Industrial Area (CVIA) smart farming clusters.
Kelantan	Rural digital entrepreneur empowerment frameworks via Digital Economy Centers (PEDi) and MDEC merchant onboarding.